

COMPUTER SCIENCE TRIPOS Part IB – 2024 – Paper 5

2 Computer Networking (awm22)

Data-Link layer  
and specific issues  
focussed upon  
the interfacing of  
data between  
physical and  
data-link layers

- (a) Physical line coding, for fiber or copper, discusses the design of systems that transform digital communications into an analog channel; the analog channel is often referred to as the physical layer.
- (i) Compare and contrast the two physical line coding systems: a pure scrambling system, and a pure block codec. [6 marks]

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*Answer:* objective: maintaining a balanced number of transitions, make identifying & recovering the bits, words, frames more-easily.

block codes are very simple and work immediately, but impose (significant) overhead (eg 25% for 8b10b)

(3 marks)

scrambler imposes probabilistic changes which can lead to long runs of zeros or ones, the shift register at the heart of a scrambler takes time to warm up (may lose data without initialization)

pure scrambling has no overhead, ensures white balance, does not allow for in-band signalling, adds decoding latency, (3 marks)

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- (ii) For each of scrambler and block code, outline the disadvantages in their sole use. Then indicate how a hybrid schema can solve such drawbacks. [4 marks]

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*Answer:*

lack of in-band control (scramblers) and high overhead (block) is attended by using a scrambler within a block but keeping the block overheads very low (1%) (1mark each)

The hybrid scheme we look at is used by 10Gbps

scramblers ensure many transitions, white balance and the control blocks can carry rich information, or even combinations of data and control info. hybridization keeps the block inefficiency low but still keeps its benefits. (2 marks for at least 2 valid reasons)

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- (b) Wireless networking faces several challenges due to the physics of radio communications. Explain, with the use of illustration, two distinct collision-detection issues unique to wireless networks, and then describe a protocol-centered approach that will overcome each issue. [10 marks]

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*Answer:* The lack of a "carrier sense" equivalent in wireless is intentionally omitted - as this rather gives the game away.

I'm expecting two parts:

an enumeration of the hidden terminal problem (making carrier sense ineffective) (2 marks)

and the exposed terminal problem where carrier sense would be overly aggressive. (2 marks)

illustrations are 2 marks each

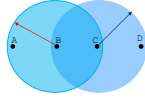
I then expect at least a discussion. using CTS/RTS to avoid needing an implicit collision detection. (2 marks)

## — *Solution notes* —

as well as potentially a speculative mechanic for preventing collisions entirely such as multiple radios a note about the inability for a single radio to both send and receive will receive an extra mark. (up to 2 marks)

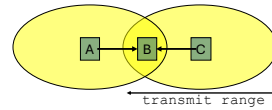
Illustrations will be variations on these (from the lecture slides)

### Exposed Terminals



- **Exposed node:** B sends a packet to A; C hears this and decides not to send a packet to D (despite the fact that this will not cause interference)!
- Carrier sense would prevent a successful transmission.

### Hidden Terminals



- A and C can both send to B but **can't hear each other**
  - A is a *hidden terminal* for C and vice versa
- Carrier Sense will be **ineffective**